Shemetov Daniil Olegovich

1. *About Me:* I am a 20-year-old , 3rd-year Software Engineering student at Lobachevsky University (UNN), Institute of ITMM. My cumulative GPA is *4.3/5.0*

2. *Hard Skills*:

- **Languages**: C++20 (Advanced), C, Go (Beginner);
- **Core Competencies**: Algorithms & Data Structures, OOP, Computer Architecture, OS Internals, Networking;
- **Testing & Quality**: Unit Testing (GTest), Clang-format, pre-commit hooks;
- **Graphics API:** DirectX 12 (D3D12)
- **Concurrency & Parallelism**: Multi-threading (std::thread/jthread), Intel Threading Building Blocks (TBB), OpenMP, MPI;
- **Tools & Environment**: CMake, Git, PowerShell/Bash scripting, Windows, Linux;
- **Console & Scripting**: PowerShell + script.ps1, Bash + script.sh;
- **Languages:** English (C1 Certified), Russian (Native)

3. *Project Experience & Skills*:

- **Parallel Programming**: As part of my coursework, I implemented high-performance multi-threaded solutions (e.g., image processing, Gaussian filters). I managed the full development pipeline: from algorithmic problem-solving and optimization to unit testing and code quality maintenance using **Git**, **CMake**, **clang-format**, and **pre-commit** hooks.

  Project Examples – https://github.com/7RosenRot/MPI_implementation (detailed technical reports available in **ReadMe.md** files)

- **Core Systems Knowledge**: Over three semesters, I completed intensive courses in **Computer Architecture**, **Operating Systems**, and **Computer Networks**, gaining a comprehensive understanding of the interaction between hardware and software components.

  Relevant Expertise: I understand low-level OS program execution, including instruction fetch, CPU core allocation, and synchronization of shared resources in critical sections. I am also familiar with client-server interactions and distributed computing implementation.

- **Graphics Pet-Project**: Currently developing a forward-renderer written in **C++20**, utilizing the **DirectX 12 API** for low-level GPU interaction. I use **CMake** for project assembly and **DebugView** for real-time diagnostic monitoring.

  Project Link – https://github.com/7RosenRot/DirectX12Engine

- **Data Structures**: To master manual memory management and container logic, I am independently re-implementing core components of the C++ Standard Library (STL).

  Successfully implemented: LinkedList, String.

  In progress: Vector.

  Future Goals: Implementing custom smart pointers – unique_ptr, shared_ptr + weak_ptr).

  Project Link – https://github.com/7RosenRot/data_structures

4. *Experience & Activities*:

- Completed intensive peer-to-peer programming training at School 21;
- Participated in the ITMO Mega-Olympiad;

5. *Soft Skills*:

- I have served as a **Team Lead** for several small-scale projects. I focus on finding a personalized approach for every team member, understanding their unique motivations, and channeling their energy toward achieving project milestones.
- I take a disciplined approach to deadlines and am always ready to support my team in challenging situations to ensure a successful shared outcome.

6. *Contacts:*

- Phone: +79200274797
- Telegram: t.me/Shemetov_Daniil
- Email: lego.den.2005@mail.ru
- GitHub: https://github.com/7RosenRot